

Name:

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Section:

ENTHALPY OF REACTION AND HESS'S LAW LAB WORKSHEET

Show all work (except average and standard deviation). Turn in the post-lab assignment before leaving the lab. Reminder: never round values prematurely; only round to the correct number of significant figures when reporting final answers and use unrounded values for subsequent calculations.

1. Data analysis

- (a) Write an equation for the conservation of energy in the calorimeter for (i) Part 0 (calibrating the calorimeter) and (ii) Part 1 (heat of solution for NaOH), using the variable names $q_{\text{calorimeter}}$, q_{cold} , q_{hot} , q_{solution} , and q_{reaction} .

(i)

(ii)

- (b) From your Part 0 data, calculate the value of the calorimeter constant C_{cal} in J/K.

$C_{\text{cal}} =$

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- (c) Calculate the molar enthalpy of reaction ΔH_{rxn} in kJ/mol for each reaction in Parts 1 - 4 below. Since many of the calculations will be similar, the feedback you receive on each Part will focus on different components of the calculation: (1) constants, units, and how each q is calculated; (2) masses and ΔT for each q ; (3) densities for masses of surroundings; (4) number of equivalent moles of reaction n_{rxn} .

(i) Part 1

(ii) Part 2

(i) $\Delta H_{\text{rxn } 1} =$

(ii) $\Delta H_{\text{rxn } 2} =$

(iii) Part 3

(iv) Part 4

(i) $\Delta H_{\text{rxn } 3} =$

(ii) $\Delta H_{\text{rxn } 4} =$

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2. Comparing results to other scholarly work

For each of the following, show an appropriate quantitative comparison and explain whether the values are consistent with each other in no more than 1 sentence.

- (a) Compare the enthalpy of solution for KNO₃ you experimentally determined in Part 2 with the literature value from the introductory material, 34.9 kJ/mol.
- (b) Use the standard heats of formation (in your textbook) to compute theoretical value for the enthalpy of reaction for reaction (3). Note: the enthalpy of formation of hydronium ion is the same as the enthalpy of formation of liquid water. Compare this theoretical value against the value you determined experimentally in Part 3.
- (c) Use your average experimental values for reactions (1) and (3), and Hess's Law, to predict the molar enthalpy of reaction (4). Compare this predicted value against the value you determined experimentally in Part 4.

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- (d) Compare your predicted molar enthalpy of reaction for reaction (4) to the literature value -100.3 kJ/mol .

3. Questions for thought

- (a) In reactions (3) and (4), the reactants were not mixed in a 1:1 ratio. What is the advantage of doing the reaction in this way rather than using equal amounts of the two reactants? Explain briefly.
- (b) A student decides to repeat reaction (3). This time, instead of using the strong acid HCl, they used another strong acid HNO_3 (nitric acid) of the same concentration. All of the other volumes and concentrations were kept the same. What effect would this have on the results of the experiment? Explain briefly.